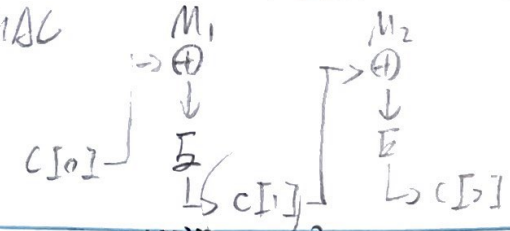
CBC-MAC



Quiz 4

$D \in \{0,1\}^{128n}$

$T: \{0,1\}^{128} \times D \to \{0,1\}^{128}$.

$K: \{0,1\}^{128}$, $M \in D$

Pare A: Difference: ECBC-MAC use difference key for AES(* T'). But T use the same key on AES(T')

The difference is on the finish step of T

Pare B: $\underline{h_1} \| \underline{C} \oplus \underline{AES^{-1}(C)}$.

① $h_1 \Rightarrow T_K(h_1) = C$.

② $\underbrace{h_1 \| AES^{-1}(C) \oplus AES^{-1}(AES^{-1}(C))}_{B.}$

A:

① $h \xleftarrow{\$} \{0,1\}^{128}$.

② $C \longleftarrow T_K(h)$

③ $B \longleftarrow AES_K^{-1}(C)$

④ $M \longleftarrow h \| B \oplus AES_K^{-1}(B)$

⑤ return (M, C)

Adv: $A_T^{MAC}(A) = Pr[(T_K(M) = C ; M \notin S ; M \in D) = 1]$

$= 1$

$M = h \| B \oplus AES_K^{-1}(B)$, $B = AES_K^{-1}(T_K(h))$

$h \in \{0,1\}^{128} \in D$

$T_K(h) \in \{0,1\}^{128} \in D$

$B \oplus AES_K^{-1}(B) \in \{0,1\}^{128} = D$.

$\underline{|M| = 256 \in D}$

$$\mathcal{A} \xleftarrow{\$} \{0,1\}^{128}.$$

$$C \longleftarrow T_K(\mathcal{A})$$

$$C = AES_K(AES_K(0^{128} \oplus \mathcal{A}))$$

$$B = AES_K^{-1}(C) = \underline{AES_K(\mathcal{A})}.$$

$$M = \mathcal{A} \| B \oplus AES_K^{-1}(B)$$
$$= \mathcal{A} \| B \oplus \mathcal{A}$$

First iteration:

$$C[1] = AES_K(0^{128} \oplus \mathcal{A}) = AES_K(\mathcal{A}) = B.$$

Second iteration:

$$\cancel{B \oplus B} \oplus \mathcal{A} = \mathcal{A}$$
$$\Downarrow$$
$$AES_K(\mathcal{A})$$
$$\Downarrow$$
$$AES_K(AES_K(\mathcal{A})) = C.$$

$$\underline{T_K(M) = C}$$

Running Time:

1 T_K query + 1 AES_K query + $O(1)$